Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		$\Delta E = hf$ and $f = c/\lambda$ (h = Planck's constant, c = speed of light) (1) energy = $hc/\lambda = 6.63 \times 10^{-34} \times 3.00 \times 10^{8} / 656 \times 10^{-9}$ (1) energy = 3.03×10^{-19} J (1)		3		3	3	
		(ii)		measuring the convergent frequency / wavelength at the convergence limit (1) calculate the ionisation energy using $\Delta E = hf$ (1)	2			2		
	(b)	(i)		hydrogen lower value since it has a smaller nuclear charge (1) electron comes from the same shell / no extra shielding (1)	2			2		
		(ii)		award (1) for any two of following • hydrogen higher value since no shielding of outer electron / lithium has shielding of outer electron • outweighs smaller nuclear charge / greater effective nuclear charge • H outer electron closer to nucleus / lithium loses electron from higher energy sub-shell	2			2		

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